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Propagation Loss in *GaN*-based Ridge Waveguides

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Abstract: *GaN* ridge waveguides were fabricated by selective area growth in an Organometallic Vapor Phase Epitaxial system. The growth enhancement on a *3.5μm* wide exposed channel versus the masked area width was measured. The propagation losses of a series of *GaN* multimode waveguides, with different widths, were measured by the out-scattering technique at $\lambda=488nm$. The internal optical loss of the *GaN* ridge waveguide was found to be $\alpha_{int} \sim 4.45cm^{-1}$. Sidewall scattering loss (α_{scat}) and the additional optical loss due to metal electrodes were also measured. The fabricated waveguides may be a basic component for integrated optic circuits.